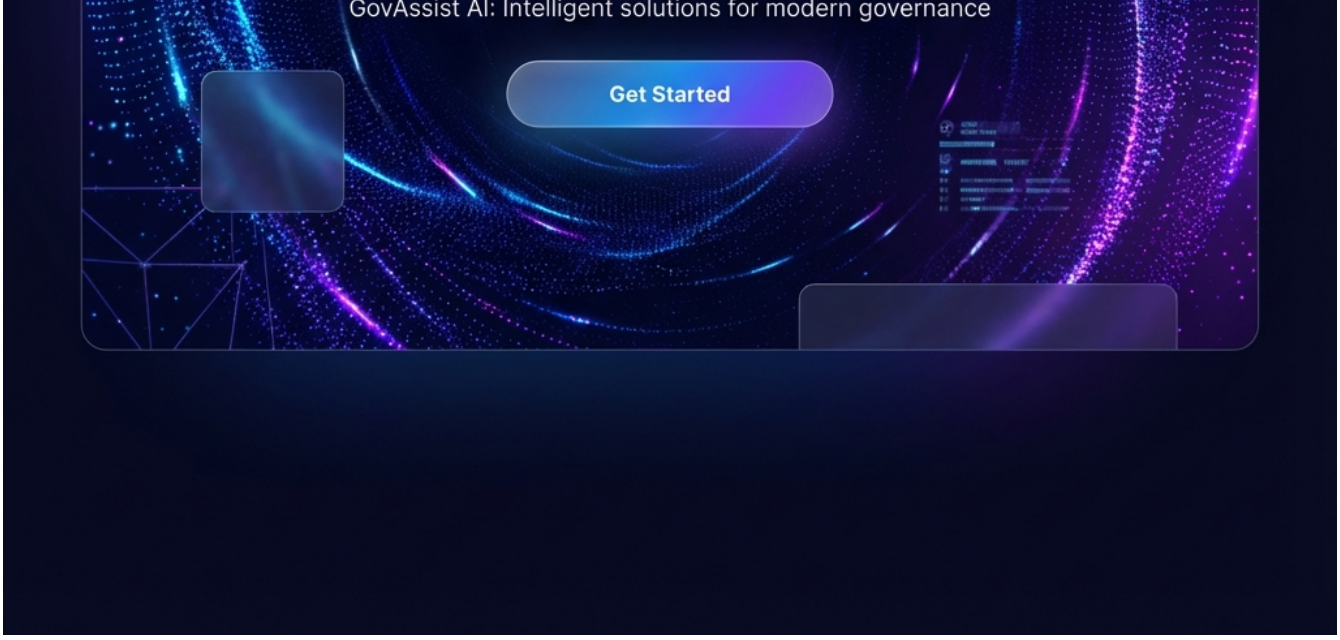

December 15, 2025

■GovAssist AI: Project Report

Project Title: GovAssist AI
Date: December 15, 2025
Version: 1.0

■1. Executive Summary

GovAssist AI is a comprehensive platform designed to bridge the gap between citizens and government welfare schemes. By leveraging advanced Artificial Intelligence (AI), including Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG), GovAssist AI simplifies the process of discovering, understanding, and applying for government schemes. The platform features a modern, accessible web interface with multi-language support (English, Hindi, Marathi), voice interaction capabilities, and document verification tools.

■2. Introduction

■2.1 Background

Government schemes are vital for social welfare, yet a significant portion of the eligible population remains unaware of or unable to access these benefits due to complex eligibility criteria, language barriers, and bureaucratic documentation processes.

■2.2 Problem Statement

- **Information Overload:** Citizens struggle to find relevant schemes among thousands of available options.
- **Language Barriers:** Most information is in English or formal government language, which is difficult for rural populations to understand.
- **Complex Documentation:** Understanding required documents and application procedures is often overwhelming.

■2.3 Objectives

- To crate a user-friendly conversational AI interface for scheme discovery.
- To provide multilingual support to reach a wider demographic.
- To enable voice interactions for users with limited literacy.
- To assist with document verification using OCR and AI analysis.

■3. System Analysis & Design

■3.1 Proposed Solution

GovAssist AI replaces the traditional search-based approach with a conversation-based approach. Users simply state their profile and needs, and the AI identifies eligible schemes, explains them in simple language, and guides them through the application process.

■3.2 Technology Stack

Frontend:
- **Framework:** React.js (Vite)
- **Styling:** Tailwind CSS (Glassmorphism Design)

- **State Management:** React Context API
- **Animations:** Framer Motion, Three.js (Particle Effects)
- **HTTP Client:** Axios

Backend:
- **Framework:** FastAPI (Python)
- **Database:** SQLite (SQLAlchemy ORM)
- **Authentication:** OAuth2 with JWT (JSON Web Tokens)
- **AI/ML:**
- **RAG Engine:** Custom implementation for context-aware responses.
- **Speech-to-Text:** OpenAI Whisper Integration.
- **OCR:** Tesseract / PDF parsing for document analysis.

■4. Key Features

■4.1 Intelligent Chatbot (RAG)
The core of GovAssist AI is a chatbot that uses Retrieval-Augmented Generation. It retrieves real-time data about schemes and generates accurate, context-aware responses, minimizing hallucinations common in standard LLMs.

■4.2 Multi-Language Support
The platform supports real-time switching between English, Hindi, and Marathi. This includes both the UI elements and the AI responses, ensuring full accessibility for regional users.

■4.3 Voice Suite & Virtual Keyboard
- **Voice Input:** Users can speak their queries using the integrated microphone. The backend transcribes this audio to text using the Whisper model.
- **Virtual Keyboard:** An on-screen keyboard allows users to type in regional languages if they lack a physical keyboard for that script.

■4.4 Document Analysis
Users can upload PDFs or images of their documents (e.g., Income Certificate, Aadhar). The system extracts text using OCR and verifies if the document is valid for specific schemes.

■4.5 Admin Dashboard
A secure admin panel allows government officials or system administrators to:
- Manage Users (RBAC).
- Approve/Reject new schemes found by the crawler.
- View system analytics.

■5. Implementation Details

■5.1 Backend Architecture
The backend is structured as a modular monolith using FastAPI.
- `app/main.py`: Entry point and application configuration.
- `app/api/`: Contains routers for different modules (Auth, Chat, Schemes, Admin).
- `app/core/`: Security, Database, and Config settings.
- `app/models/`: SQLAlchemy database models.

■5.2 Frontend Architecture

The frontend is built for performance and user experience.
- **Responsive Design:** Mobile-first approach ensuring the app works on low-end devices.
- **Lazy Loading:** Components and pages are loaded on demand to reduce initial load time.
- **Security:** Protected routes ensuring only authenticated users access sensitive features.

■6. Future Scope

- **Mobile Application:** Native Android/iOS apps for better offline capabilities.
- **Direct Integration:** Integration with government APIs for direct application submission.
- **More Languages:** Expansion to all 22 scheduled languages of India.

■7. Conclusion

GovAssist AI successfully demonstrates how technology can be used for social good. By simplifyi
ng access to information and breaking down language barriers, it empowers citizens to claim the
ir rights and benefits effectively. The system is scalable, secure, and ready for pilot deploym
ent.

References

1. FastAPI (Backend Framework): https://fastapi.tiangolo.com/
2. React & Vite (Frontend): https://react.dev/
3. OpenAI Whisper (Speech Recognition): https://github.com/openai/whisper
4. Tailwind CSS (Styling): https://tailwindcss.com/
5. SQLAlchemy (Database ORM): https://www.sqlalchemy.org/
6. Framer Motion (Animations): https://www.framer.com/motion/
7. ReportLab (PDF Generation): https://www.reportlab.com/
8. Python-PPTX (Presentation Generation): https://python-pptx.readthedocs.io/